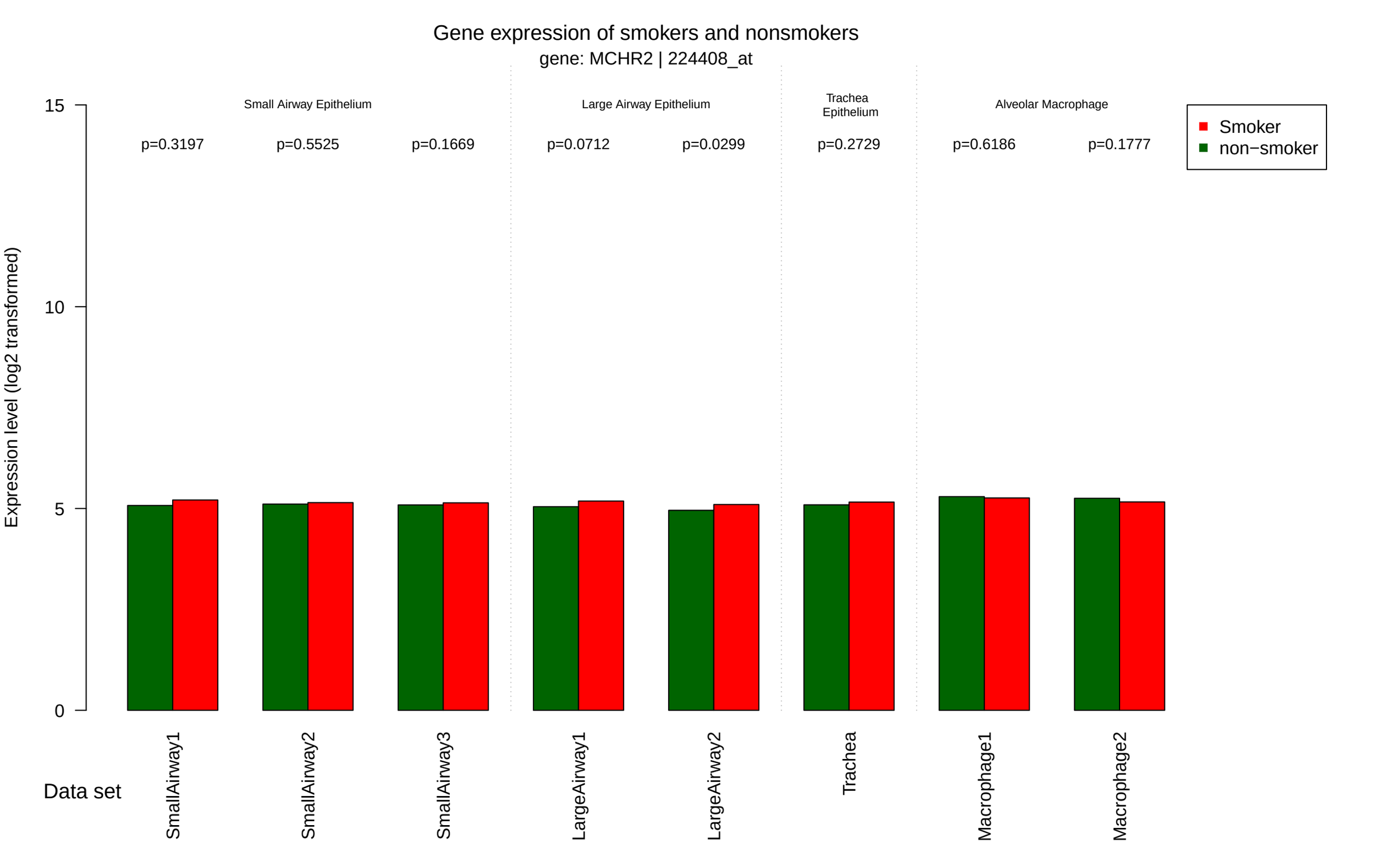
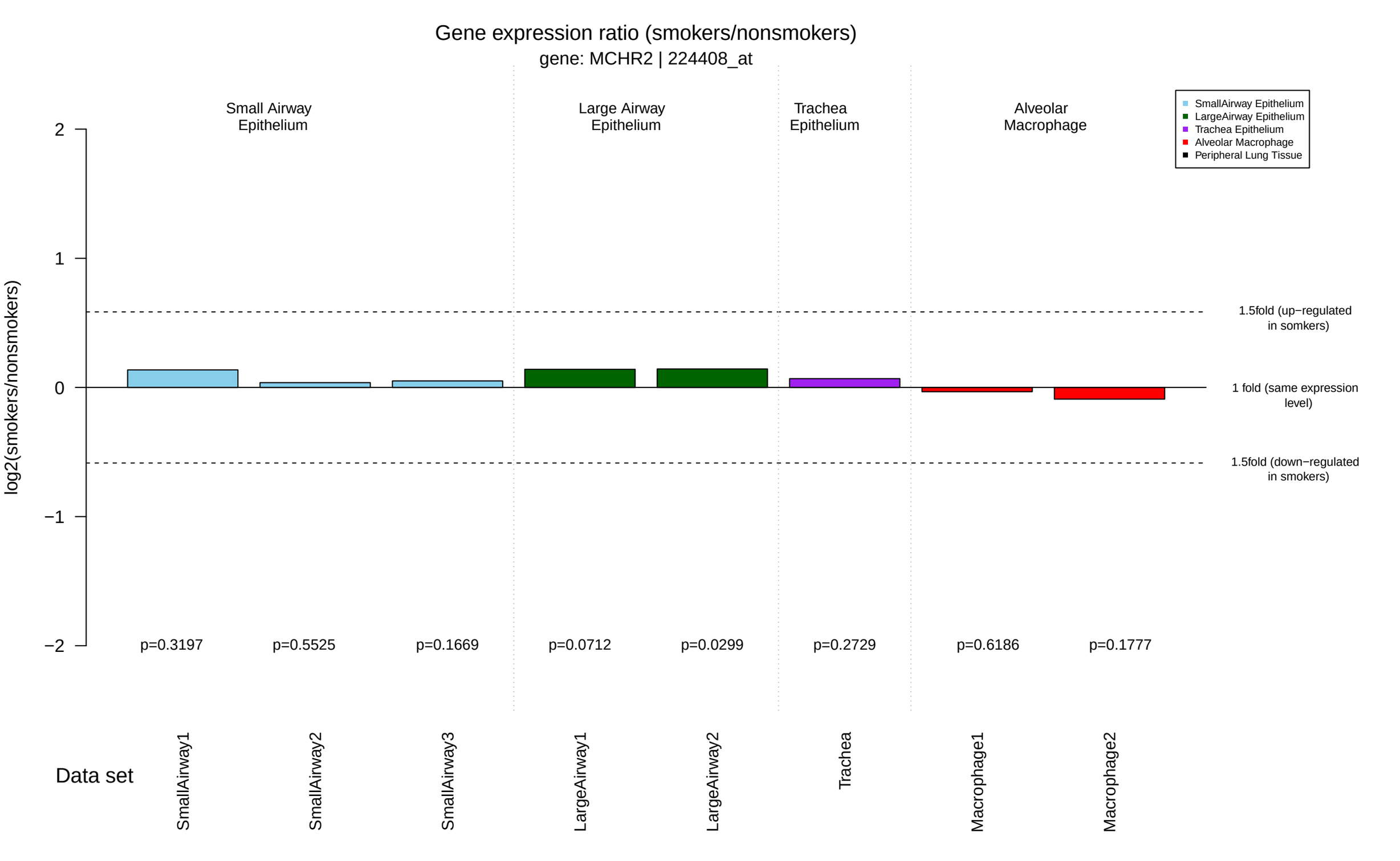
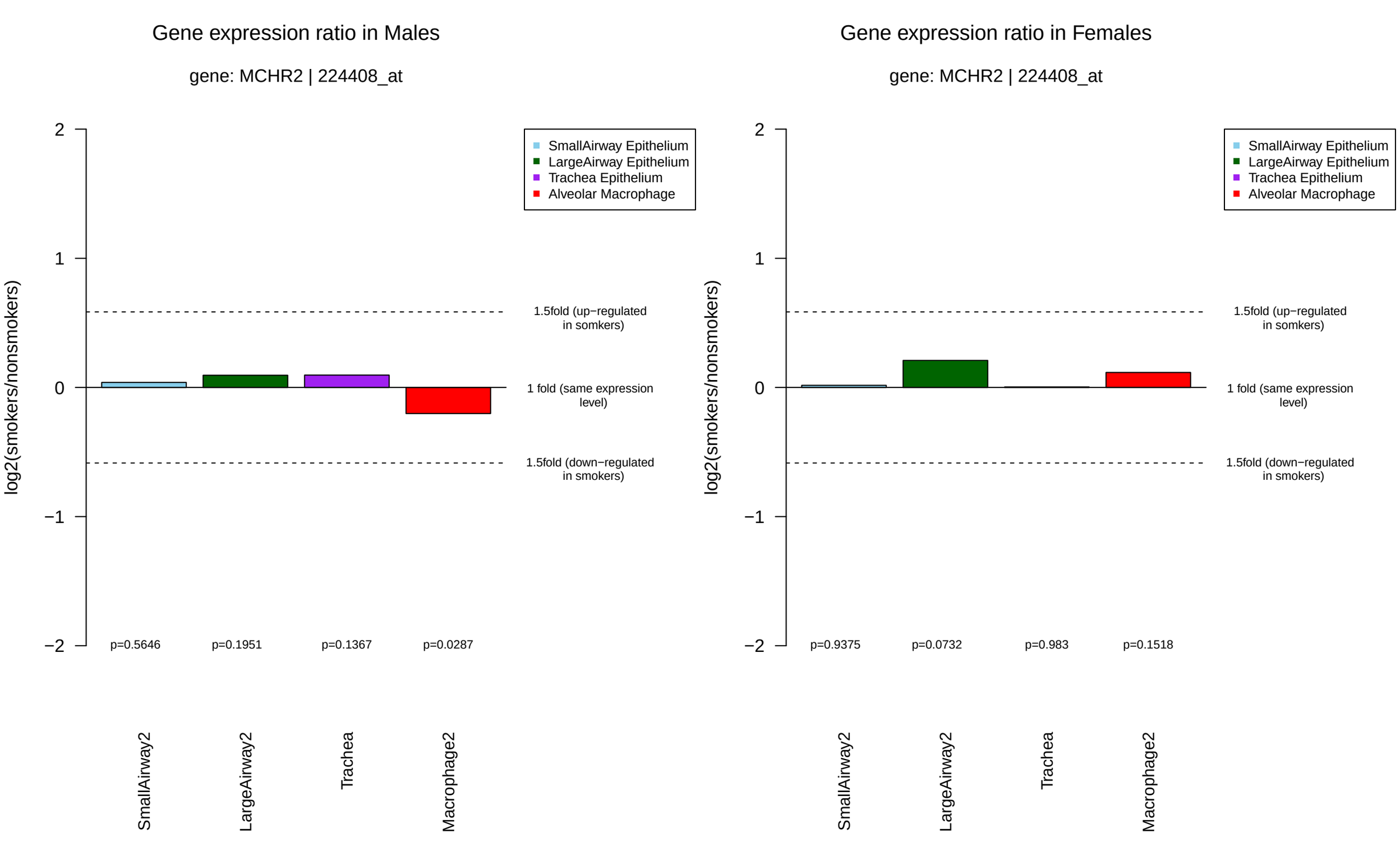
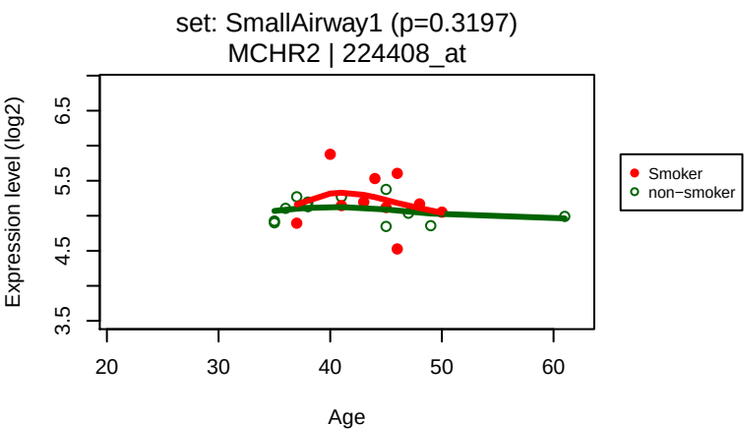
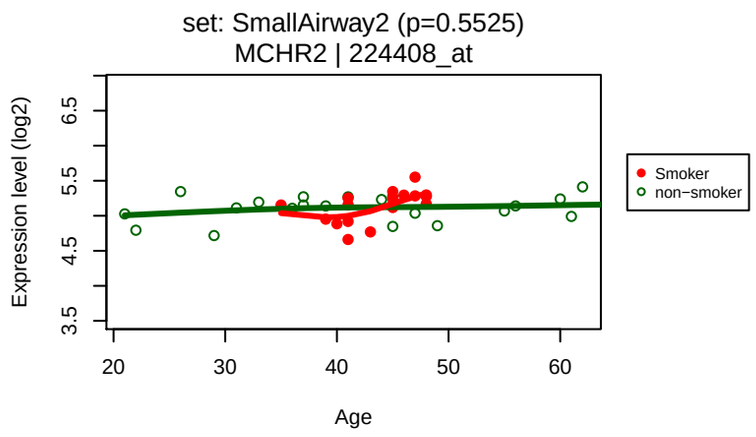
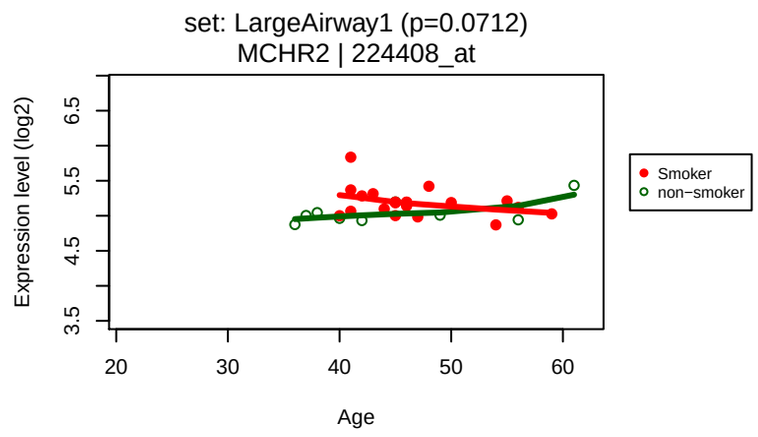

Gene expression of smokers and nonsmokers
gene: MCHR2 | 224408_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3197
p=0.5525
p=0.1669
p=0.0712
p=0.0299
p=0.2729
p=0.6186
p=0.1777
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

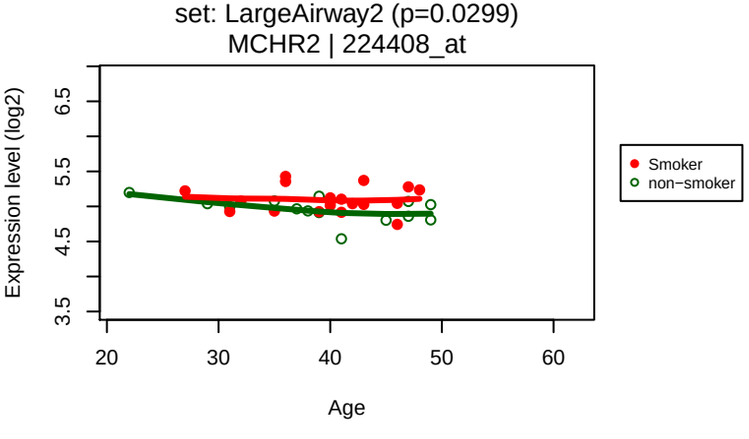

Gene expression ratio (smokers/nonsmokers)
gene: MCHR2 | 224408_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3197
p=0.5525
p=0.1669
p=0.0712
p=0.0299
p=0.2729
p=0.6186
p=0.1777
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

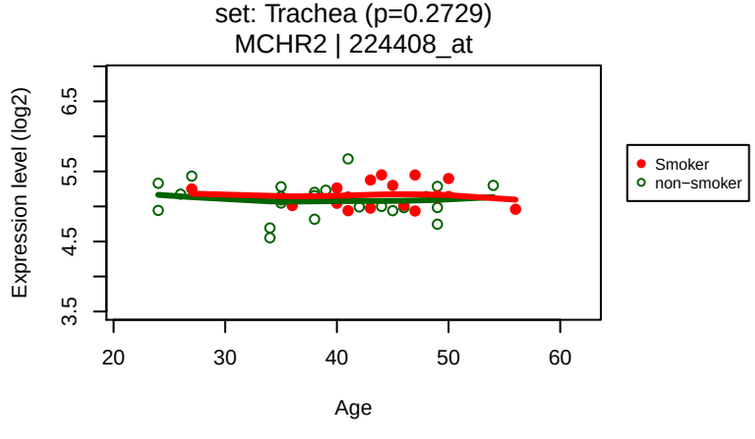

Gene expression ratio in Males
gene: MCHR2 | 224408_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5646
p=0.1951
p=0.1367
p=0.0287
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MCHR2 | 224408_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9375
p=0.0732
p=0.983
p=0.1518
SmallAirway2
LargeAirway2
Trachea
Macrophage2

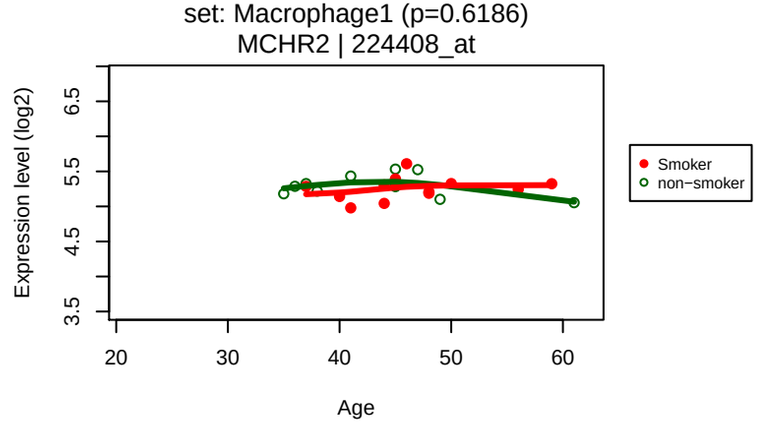

set: SmallAirway1 (p=0.3197)
MCHR2 | 224408_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker

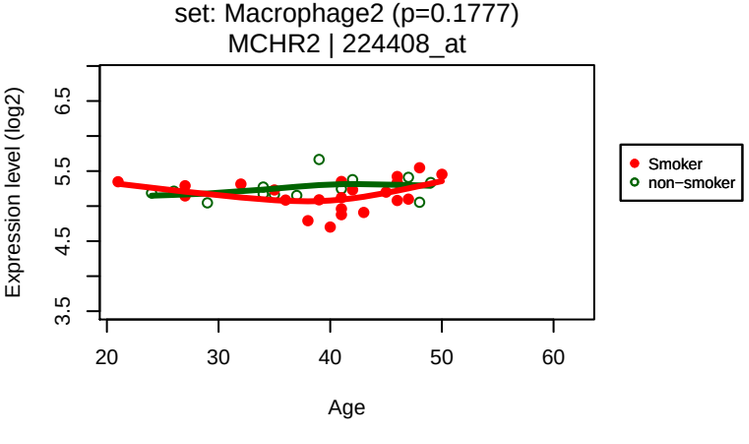

set: SmallAirway2 (p=0.5525)
MCHR2 | 224408_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker

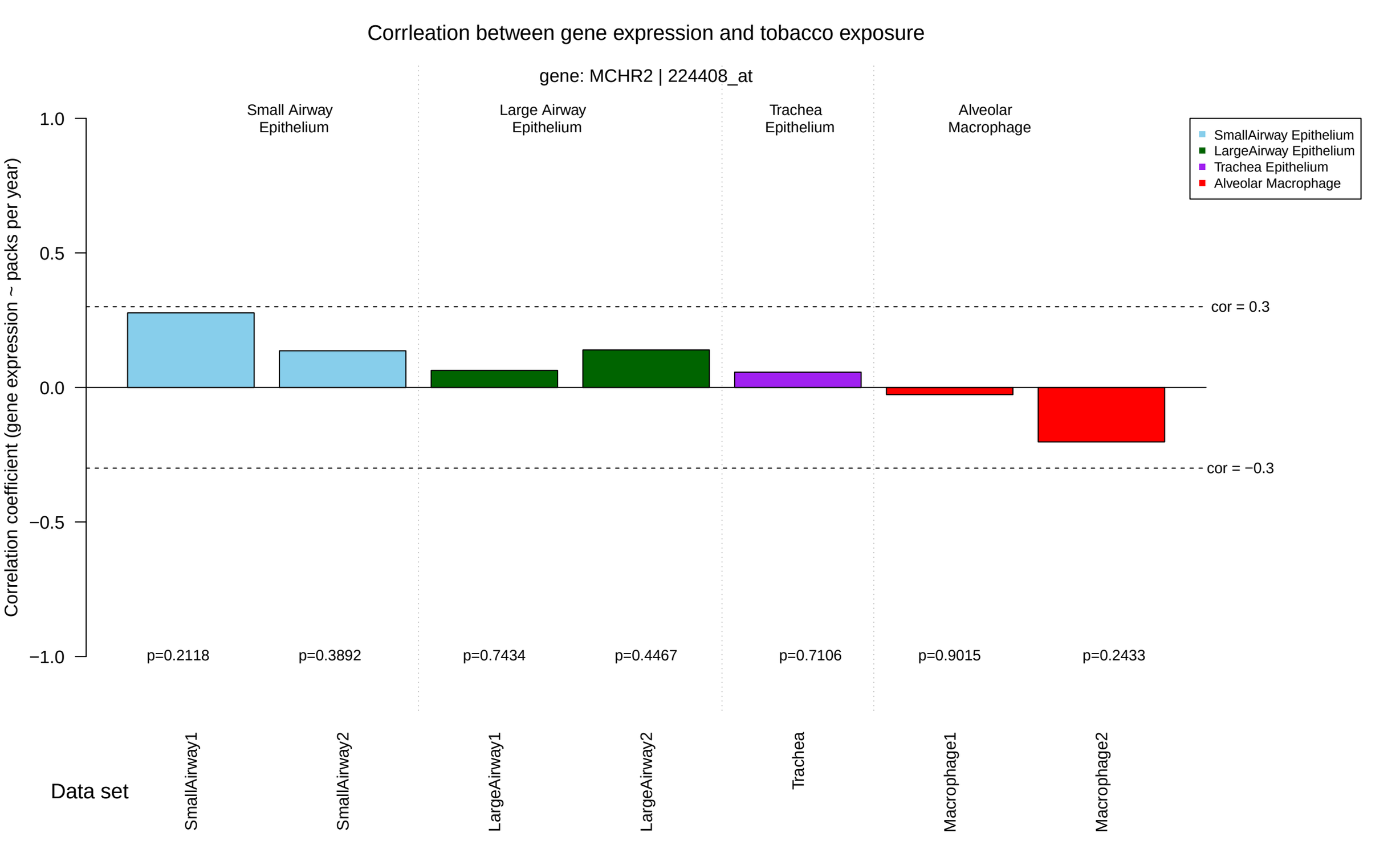

set: LargeAirway1 (p=0.0712)
MCHR2 | 224408_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0299)
MCHR2 | 224408_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.2729)
MCHR2 | 224408_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6186)
MCHR2 | 224408_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1777)
MCHR2 | 224408_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MCHR2 | 224408_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2118
p=0.3892
p=0.7434
p=0.4467
p=0.7106
p=0.9015
p=0.2433
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2